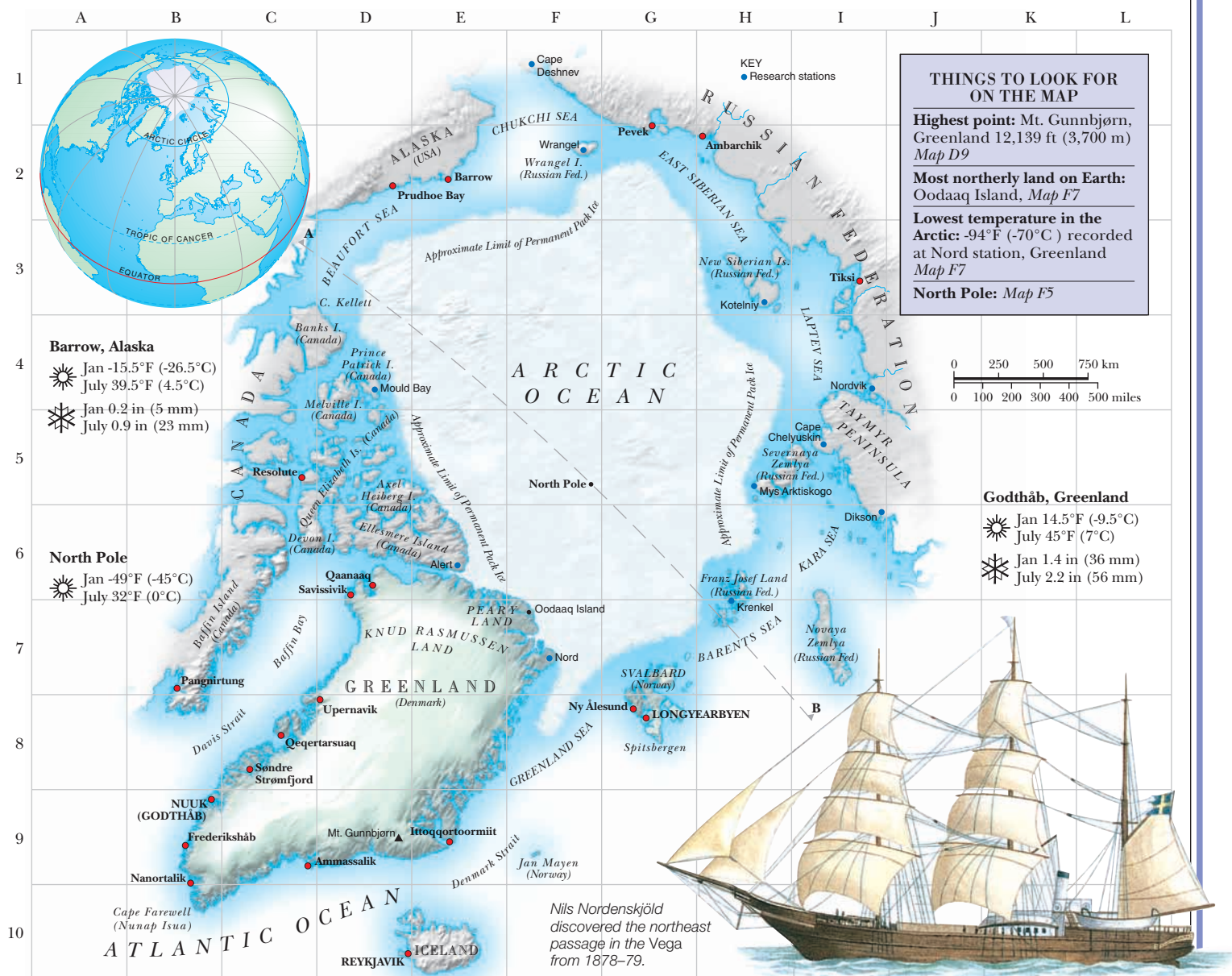




ICE BREAKERS

The Arctic sea ice is usually about 6 ft (2 m) thick. Ships called ice breakers are specially constructed with reinforced hulls and bows shaped for plowing through this icy obstacle. Their powerful engines push the bow on top of the ice until the weight of the ship breaks it, clearing a passage for other sea vessels.



ARCTIC EXPLORATION

In the 16th century, European sailors first explored the Arctic seas in search of a new trade route to Asia. English sailors began to map the northwest passage around the top of Canada, while the Dutch explored the northeast passage around Siberia. By 1906, both routes had been successfully navigated. Three years later, an American, Robert Peary, claimed to have reached the North Pole itself. However, the speed of his journey has led some explorers to doubt his achievement.



Roald Amundsen sailed through the northwest passage in the Gjöa from 1903–06.

This cross-section follows the line A–B on the map above.

CROSS-SECTION THROUGH THE ARCTIC

For centuries, some Arctic explorers believed that the polar ice lay on top of a vast continent. In 1958, an American atomic-powered submarine, the USS *Nautilus*, demonstrated that this was untrue by sailing from Alaska to Svalbard underneath the ice. Subsequent exploration beneath the ice has revealed the ridges and basins of the seabed below.

